

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Midterm Examination of Semester 2

Subject: 341152 Electric Circuits II

Date: 12 March 2015

Year: 2014

Section: 5-6

Time: 15.00-17.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken out of the examination room.
 3. Text books are **NOT** allowed.
 4. A calculator is allowed.
 5. Electronic communication devices are **NOT** allowed in the examination room.
 6. The examination has 10 pages (including this page), 3 parts (59 questions) and a total score of 75 points.
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Part 1 (25 points) True/False

Instruction: Mark the correct answer for the following questions in the provided answer sheet.

1. The time constant for a capacitor circuit equals its capacitance divided by its resistance.
2. An ideal inductor looks like an open circuit to dc current.
3. The steady-state level of the capacitor current can be found by substituting the short-circuit equivalent.
4. The current across a capacitor lags the voltage through it by 90° .
5. Capacitive reactance increases inversely to frequency.
6. Power factor of inductive circuit is equal to one.
7. For a purely resistive element, the voltage and the current through the element are in phase.
8. A capacitive network has a lagging power factor.
9. Total impedance of a system is the sum of the individual impedances.
10. Admittance is the reciprocal of impedance.
11. Capacitance is directly proportional to the distance between plates.
12. The magnitude of polar form is in average value.
13. When charging capacitor, the voltage of capacitor decreases exponentially.
14. Resistor value is dependent from frequency.
15. At very high frequency, capacitor acts as an open circuit.
16. When closing switch of RL in DC circuit, electric current can flow in right at the beginning.
17. When charging capacitor, the current of capacitor increases exponentially.

18. The unit of inductive reactance is in Ohm.
19. The unit of active power is in VA.
20. The unit of power factor is in radian.
21. The current divider rule for dc circuits can be used for ac circuits.
22. For a series ac circuit with reactive elements, the current in coils or capacitors are always in the same direction on a phasor diagram.
23. In RC series circuit, electric current leads the voltage at power supply by 90 degrees.
24. In parallel network, total admittance is the sum of the individual admittance.
25. The heavier the current flow for a given voltage, the larger the value for admittance.

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6. Find the capacitance of a parallel-plate air dielectric capacitor if the area of each of its plates is 0.1 m^2 and the distance between the plates is 2.0 mm .
 (a) 1.44 nF (b) 0.62 nF (c) 0.44 nF (d) 0.14 nF
7. Given the voltage $v_c = 12 \text{ V}(1 - e^{-t/100 \mu\text{s}})$. What is the time constant in this circuit?
 (a) 100 s (b) $12 \mu\text{s}$ (c) $500 \mu\text{s}$ (d) $100 \mu\text{s}$
8. From Problem 7, what is the voltage at $t = 1 \text{ ms}$?
 (a) 12 V (b) 0.12 V (c) 7.58 V (d) 0.99 V

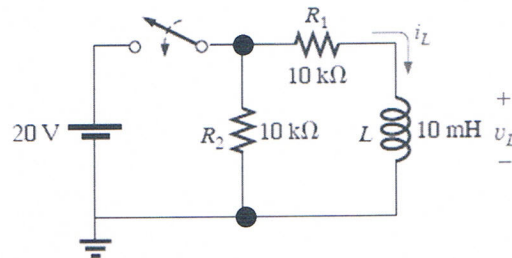


Figure 3

9. See Figure 3. After closing the switch, what will the current (i_L) be after the circuit voltages and currents have reached steady-state values? Assume that the inductor is an ideal (lossless) device.
 (a) 0 A (b) 2 mA (c) 4 mA (d) 10 mA
10. The currents and voltages are in steady state. Find V_1

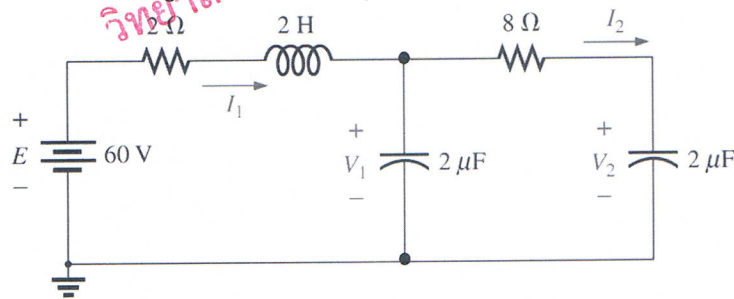


Figure 4

- (a) 60 V (b) 48 V (c) 12 V (d) 0 V
11. See Figure 4. Find the current I_1 ?
 (a) 0 A (b) 6 A (c) 7.5 A (d) 30 A
12. What is the capacitive reactance at 0.5 Hz of a 1 F capacitor?
 (a) $\pi \Omega$ (b) $2\pi \Omega$ (c) $1/\pi \Omega$ (d) $1/2\pi \Omega$
13. At what frequency does a $10/\pi \text{ mH}$ inductor have a reactance of 1000Ω ?
 (a) 50 Hz (b) 500 Hz (c) 50 kHz (d) 5 MHz

14. If the voltage $v=50\sin(500t)$ is impressed across a $25\ \Omega$ resistor, which equation describes the resistor current?
- (a) $2\sin(500t + 90^\circ)$ (b) $2\sin(500t - 90^\circ)$
 (c) $2\sin(500t)$ (d) $2\sin(50t)$
15. The voltage across a 100 mH coil is $i=100\sin(20t)$. Which of these expressions describes the voltage?
- (a) $100\sin(20t - 90^\circ)$ (b) $200\sin(20t + 90^\circ)$
 (c) $200\sin(20t)$ (d) $200\sin(20t - 90^\circ)$
16. Which one of the following polar values is equivalent to $30 + j40$?
- (a) $50 \angle 36.9^\circ$ (b) $50 \angle 53.1^\circ$ (c) $50 \angle -36.9^\circ$ (d) $50 \angle -53.1^\circ$
17. Which one of the following values is equivalent to j^{10} ?
- (a) $1 \angle 0^\circ$ (b) $1 \angle 90^\circ$ (c) $1 \angle 180^\circ$ (d) $1 \angle -90^\circ$
18. Which one of the following values is equivalent to $1 \angle 180^\circ$?
- (a) $-1 \angle 0^\circ$ (b) $1 \angle 90^\circ$ (c) $1 \angle 0^\circ$ (d) $1 \angle -90^\circ$
19. Which one of the following rectangular values is equal to the polar form $20 \angle -55^\circ$?
- (a) $16.38 - j11.47$ (b) $16.38 + j11.47$
 (c) $11.47 - j16.38$ (d) $11.47 + j16.38$
20. Which one of the following phasor domain expressions is equivalent to the time domain expression of $50\sin(\omega t - 15^\circ)$?
- (a) $35.35 \angle -15^\circ$ (b) $50 \angle 15^\circ$
 (c) $35.35 \angle 15^\circ$ (d) $70.7 \angle -15^\circ$

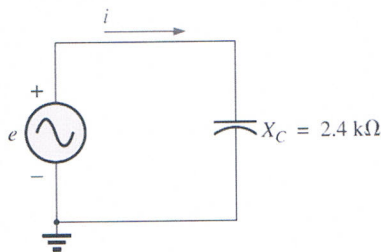
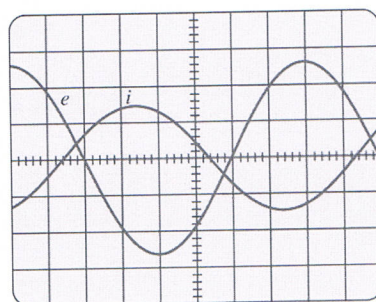


Figure 5

21. See Figure 5. If $e = 150 \sin(157t + 30^\circ)$, what is the current i ?
- (a) $62.5\text{ A} \sin(157t + 120^\circ)$ (b) $62.5\text{ A} \sin(157t - 60^\circ)$
 (c) $62.5\text{ mA} \sin(157t - 60^\circ)$ (d) $62.5\text{ mA} \sin(157t + 120^\circ)$



Vertical sensitivity = 0.5 V/div.
Horizontal sensitivity = 1 ms/div.

Figure 6

22. See Figure 6. What relationship exists between voltages e and i ?
 (a) e lags i by 207° (b) e leads i by 27°
 (c) e lags i by 27° (d) e leads i by 207°
23. See Figure 6. What is the average value of the current i ?
 (a) 1.5 A (b) 1.25 A (c) 0.75 A (d) 0 A
24. See Figure 6. What is the frequency of the waveforms?
 (a) 100 Hz (b) 125 Hz (c) 130 Hz (d) 150 Hz
25. See Figure 6. What is the RMS value of the voltage e ?
 (a) 1.25 V (b) 0.88 V (c) 1.77 V (d) 2.5 V
26. See Figure 6. What is the V_{p-p} value of the current i ?
 (a) 0.75 A (b) 1.25 A (c) 1.5 A (d) 2.5 A
27. What is the phase relationship between the waveform $v = 2 \sin(140t + 27^\circ)$ and $i = 19 \sin(140t - 43^\circ)$?
 (a) v leads i by 70° (b) i leads v by 55° .
 (c) i leads v by 70° (d) v leads i by 55°
28. Find the RMS value of the waveform in Figure 7.

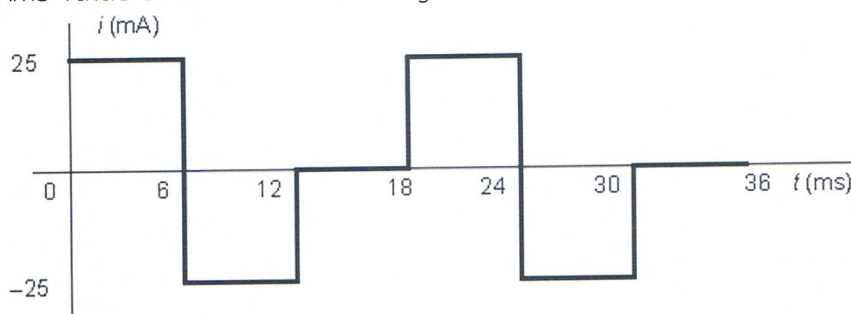


Figure 7

- (a) 50 mA (b) 25 mA (c) 16.67 mA (d) 20.41 mA
29. Using the waveform in Figure 7, find the frequency of the current.
 (a) 83.3 Hz (b) 167 Hz (c) 55.6 Hz (d) 0.083 Hz
30. Evaluate the expression $(4 + j5) / (39 \angle -42^\circ)$ and express the answer in polar form.
 (a) $0.16 \angle 9.3^\circ$ (b) $0.16 \angle 93.34^\circ$ (c) $249.6 \angle 93.34^\circ$ (d) $249.6 \angle 9.3^\circ$

Part 3 (20 points) Calculation

Instruction: Shows the mathematical expressions and answers of following problems.

1. Find the total ADMITTANCE (Y_T) and IMPEDANCE (Z_T) in the circuit of Figure 8.

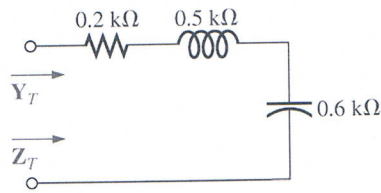
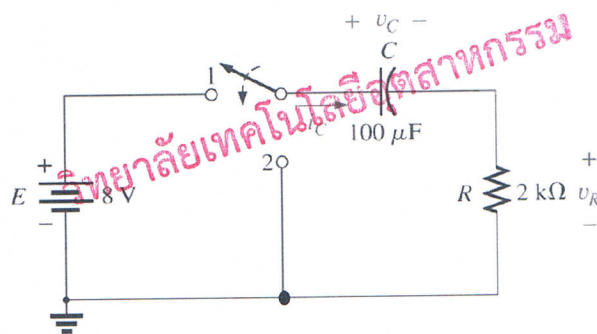
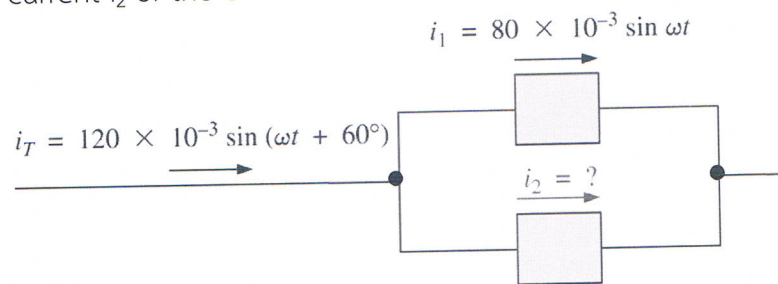


Figure 8

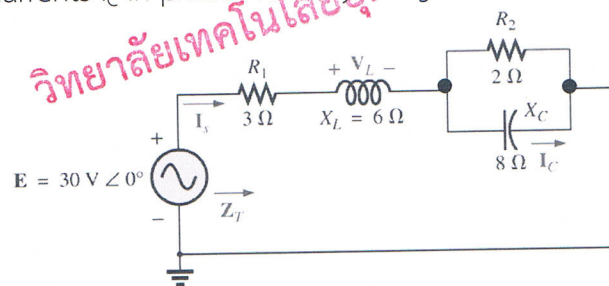
2. From the circuit below, find
- When the switch is in position 1, find time constant and write the mathematical expression of i_C and v_C ,
 - When the switch is in position 2, write the mathematical expression of i_C and v_C



3. Find the current i_2 of the circuit.



4. Calculate the currents I_C in phasor form by using the current divider rule.



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Answer Sheet

Name: _____ ID: _____

Subject: 341152 Electric Circuits II

Part 1

	True	False
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Part 2

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